

Smart Study Guide: Class 12 Chemistry

Analysis of Priority Sets: 56/5/1, 56/7/1, and HFG1E/5

Created by Pragya • Optimized for High Recovery

1. THE "GUARANTEED MARKS" LIST (HIGH PROBABILITY)

These topics appeared in **all three papers**. Mastering these ensures you cover the core weightage of the exam.

Branch	Chapter	Top Priority Topic	Typical Question Pattern
Physical	Electro-chemistry	Nernst Equation	Numerical: Calculate emf (E_{cell}) and ΔG° for a cell reaction (e.g., Zn/Cu or Mg/Cu cells) at 298K.
		Kohlrausch Law	Numerical: Calculate limiting molar conductivity (Λ_m°) or degree of dissociation (α) given conductivity (κ).
Physical	Solutions	Colligative Properties & i factor	Numerical: Mass of solute for freezing/boiling point. Watch for Van't Hoff factor (i) for salts like MgCl_2 .
Physical	Kinetics	First Order Kinetics	Proof: Show time for 99% is twice of 90%. Numerical: Calculate k or t .
Inorganic	Coordination	VBT & CFT	Reasoning: Color (d-d transition). Compare geometry/magnetic nature (e.g., $[\text{Ni}(\text{CN})_4]^{2-}$ vs $[\text{Ni}(\text{Cl})_4]^{2-}$).
Organic	Haloalkanes	SN1 vs SN2	Reasoning: Arrange compounds in increasing order of reactivity. Focus on steric hindrance and carbocation stability.
Organic	Aldehydes / Ketones	Name Reactions	Direct/Complete: Cannizzaro Reaction (repeated in all sets), Sandmeyer's, Clemmensen.
Organic	Amines	Basicity Order	Reasoning: Arrange amines in aqueous phase vs gas phase (steric/solvation effects).

2. CHAPTER-WISE IMPORTANT QUESTIONS (DIRECT REPEATS)

Physical Chemistry

- **Chemical Kinetics:** "The rate of a reaction quadruples when the temperature changes from 300 K to 320 K. Calculate the energy of activation (E_a)."
- **Chemical Kinetics:** Plot graphs for zero and first-order reactions (Concentration vs Time).
- **Electrochemistry:** "State Faraday's first law of electrolysis."
- **Electrochemistry:** Explain the variation of conductivity and molar conductivity with concentration.

Inorganic Chemistry

- **d- and f-Block:** "Why is E° (Mn^{2+}/Mn) highly negative?" or "Why is Mn^{3+} a strong oxidizing agent?"
- **d- and f-Block:** "Explain the cause and consequences of Lanthanoid Contraction (e.g., Zr and Hf have similar radii)."
- **d- and f-Block:** Preparation of $K_2Cr_2O_7$ or $KMnO_4$ from ores.
- **Coordination:** Define "Ambidentate ligand" or "Linkage Isomerism" (e.g., NO_2 vs ONO ligands).

Organic Chemistry

- **Distinction Tests (2-3 Marks):**
 - **Iodoform Test:** Methyl Ketones (CH_3-CO-) or Methyl Carbinols.
 - **Tollens'/Fehling's:** Aldehydes (pos) vs Ketones (neg).
 - **Hinsberg Test:** Distinguish 1° , 2° , 3° amines.
- **Biomolecules:** Structure: "Write the reaction of D-Glucose with HCN , Br_2 water, or HI ." (Confirms carbonyl, aldehyde, and straight chain).
- **Biomolecules:** Definitions: Peptide linkage, Denaturation of proteins, Glycosidic linkage.

3. EASY-TO-REVISE STUDY FORMAT

Section A: The Formulae & Definitions (Quick Marks)

- ☐ **Nernst Equation:** $E_{\text{cell}} = E^{\circ}_{\text{cell}} - (0.059/n) \log Q$
- ☐ **First Order Rate Law:** $k = (2.303/t) \log ([R]_0/[R])$
- ☐ **Arrhenius Equation:** $\log (k_2/k_1) = (E_a / 2.303 R) [(T_2-T_1)/(T_1T_2)]$
- ☐ **Spin-only Magnetic Moment:** $\mu = \sqrt{n(n+2)} \text{ BM}$

Section B: The "Account For" Reasoning List

- ☐ Why transition metals are colored/paramagnetic/act as catalysts.
- ☐ Why Phenol is more acidic than alcohol.
- ☐ Why Aniline does not undergo Friedel-Crafts reaction.
- ☐ Why H_3PO_3 is diprotic (structure based).

Section C: Name Reactions & Tests

- ☐ **Reactions:** Cannizzaro, Aldol, Rosenmund, Stephen, Hoffmann Bromamide.
- ☐ **Tests:** Lucas test (Alcohols), Iodoform (Methyl Ketones), Carbylamine (Primary Amines).

Section D: Case Studies

- ☐ **Focus Areas:** Biomolecules (DNA/Proteins) and Solutions/Electrochemistry (Batteries/Fuel Cells).